

EXPERIENCE

Microsoft

Senior Software Engineer

2022 – Present

Fabric Spark Team

- Built [Fabric](#) Spark Coding Agent from 0 to 1: code generation, error-fix, and perf optimization for notebooks; live in 51 countries/regions.
 - Led the agent system architecture: prompt orchestration, tool-calling loop, front/back-end UX tool protocol, tool registry, and multi-turn execution.
 - Built a Spark-domain RAG knowledge base and wrapped diagnostic services like History Server as MCP tools (Job, Stage, Executor, Driver/Livy logs), so the agent locates issues from real runtime evidence.
 - Built an Agent Eval system (code gen, error-fix, perf diagnosis, tool-calling, regression) and a pre-cooked context mechanism (e.g. `getContextByCodeCell`) aggregating cell/job/app/error context, improving diagnosis efficiency 5x and cutting token usage 90%.
 - Open-sourced core Spark Skills to [microsoft/skills-for-fabric](#) for third-party agents to reuse Fabric Spark diagnosis and tuning.
- Spark Monitoring & Diagnostics: supports 10M+ Spark applications/day.
 - Customized Spark/Hadoop for specific monitoring scenarios, e.g. read/write time and per-file read/write size.
 - Analyzes commands and code run by Apache Spark and displays real-time advice for notebook runs via [Spark Advisor](#).
 - Auto-analyzes [recurring Spark jobs](#) with duration/I/O trends, outlier detection, and root-cause attribution; published at [VLDB 2025](#).
 - Added high-concurrency sessions (users sharing one Spark app) and live sessions (pre-started to cut warm-up) with related monitoring.

JD.com

Senior Software Engineer

2020 – 2022

Advertising Data Team

- Owned JD.com's foundational advertising data.
 - Ingested real-time impression/click/spend data from on-site Jingzhuntong and off-site channels: Tencent GDT, ByteDance Ocean Engine, Baidu, Google, and Meta.
 - Built a real-time Flink pipeline processing 1B+ ad events/day with exactly-once semantics and pluggable sources/sinks at sub-30 s latency, powering analytics/BI, budget overspend control (stop-loss), and billing reconciliation.
 - Built a Customer Data Platform (CDP) with OneID identity resolution, unifying cross-channel profiles to improve targeting precision and ROI.
 - Handled 618 and Double 11 peak traffic with disaster recovery and full-link stress testing.
- 2022 CCTV Spring Festival Gala red-packet project: owned the data, tallying per-round participants/amounts in real time for stop-loss and forecasting, peak QPS 500K.

Xiaomi

Software Engineer

2018 – 2020

Data Platform Team

- Built ETL pipelines processing billions of daily events, a unified, reliable foundation for downstream analytics and recommendation.
- Built Xiaomi's video recommendation system (feature engineering, recall, ranking); introduced drag-and-drop self-service BI, cutting report delivery from days to minutes.

AskingData

Software Engineer

2016 – 2018

Data Team

- Built high-throughput crawlers collecting billions of e-commerce records daily; cracked anti-crawler challenges (IP bans, CAPTCHA, obfuscation, encryption), cleaned with Spark, plus an anti-crawler defense system.

EDUCATION

Huazhong University of Science and Technology

Bachelor 2012 – 2016